

Mid-term exam project for the electronic circuit lab.

ATX Bench Power Supply

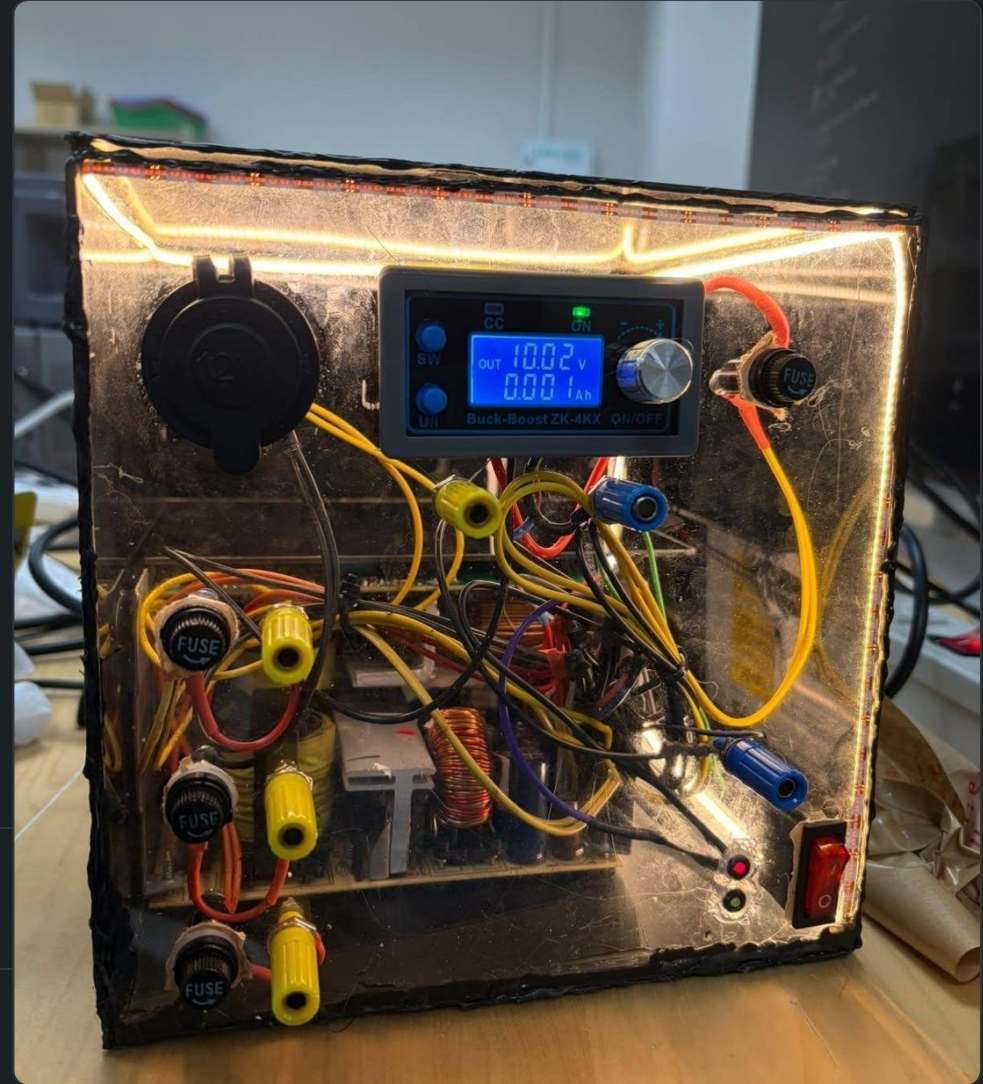
Turning a closed-case ATX unit into a safe, multi-rail bench supply — fixed rails plus one adjustable output.

Team

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Source PSU

SVOA, 200W



Project overview

A closed-case ATX unit converted into a bench supply: fixed +3.3V, +5V, +12V and -12V rails, plus one adjustable buck-boost output.

Accepted requirements

- Fused, insulated fixed rails
- Adjustable output
- Dedicated PS_ON control
- Standby and run indicators

Contributions

- Korawee: wiring, electrical design, calculations
- pandin: acrylic enclosure, design, testing

Safety boundary and source specs

Safety boundary

The metal enclosure stays closed. High-voltage internals are unmodified. Wiring changes happen only with the AC cable removed.

+3.3V

20A

+5V

20A

+12V

37.4A

+5VSB

3A

450W

combined, per PSU label

Stop conditions

Rails cycling repeatedly, voltage past $\pm 5\%$ of nominal, smoke or odor, or a blown fuse.

Rails and bill of materials

+3.3V

fixed

+5V

fixed

+12V

fixed

-12V

fixed

0.5 - 30V

added output, buck-boost, 4A

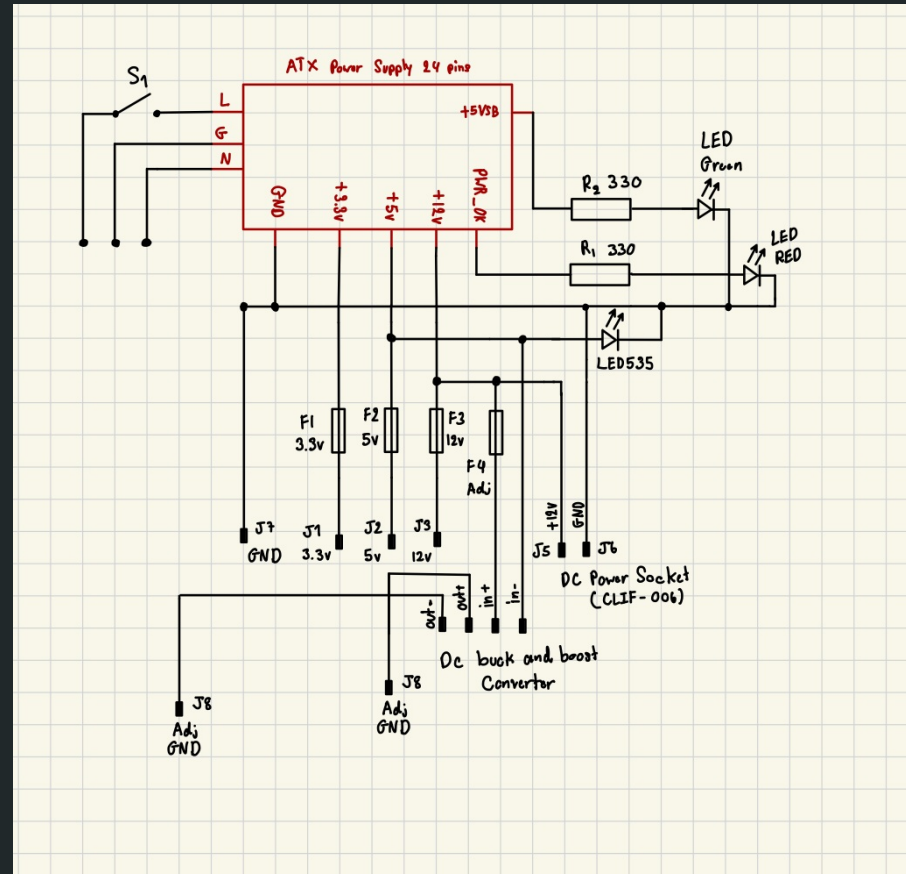
Item	Spec	THB
Acrylic enclosure	20×20×3mm, 6 pc	120
ATX PSU	200W	100
Buck-boost module	in 5-30V, out 0.5-30V, 4A	315
Fuses	10A 250V, ×4	4
Banana jacks	10A 30VDC, ×6	72
Paint, LEDs, switch, DC socket	mixed hardware	256

867 THB

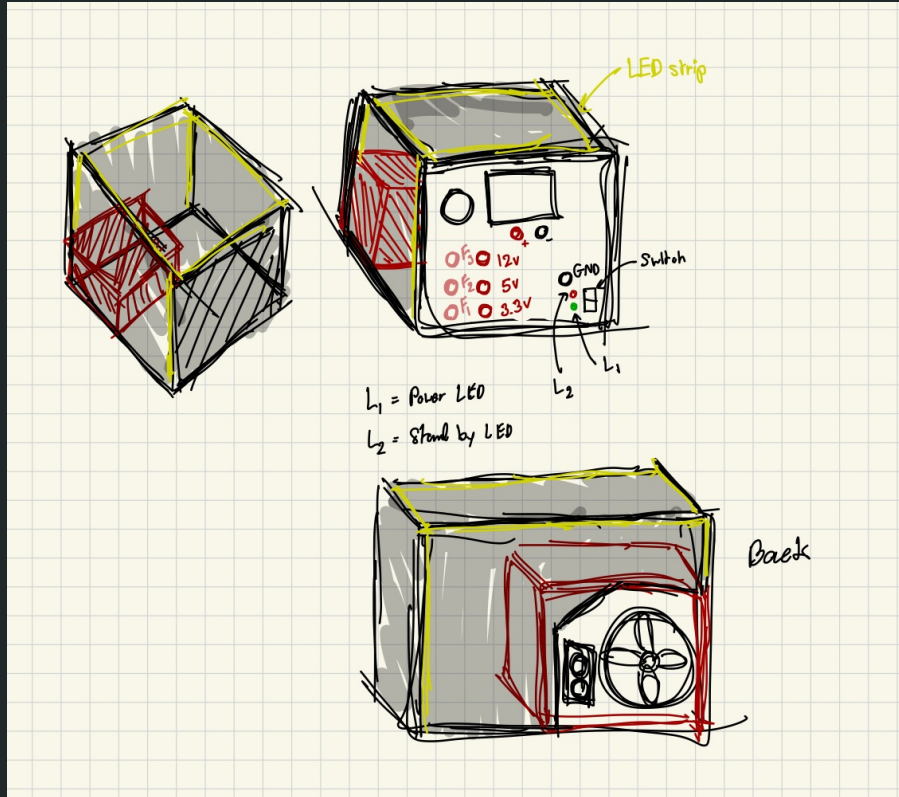
total build cost

Circuit schematic

Fixed rails run through fuses F1 to F3; the adjustable output runs through F4. Power and standby status come from PWR_OK and +5VSB.



Enclosure design



Sketch: front panel, LED strip, rear fan



Built

Engineering calculations

5A

fuse rating, margin on 18AWG's 16A

≈85%

assumed buck-boost efficiency, η

<60°C

component thermal limit

Converter and loss

$$P \approx V_{\text{out}} \times I_{\text{out}} \times \eta$$

Thermal

$$P = I^2 R, \text{ checked against full test load}$$

Testing and acceptance

Unpowered checks

No rail-to-rail shorts found from +3.3V, +5V or +12V to ground.

3.47V

+3.3VSB

5.07V

+5VSB

12.3V

+12VSB

0 - 30V

adjustable sweep, verified

38°C

peak, +12V fuse holder, 15 min at 3A

Fault, then correction

The switch was wired so the supply stayed always-on, bypassing it. Corrected by relocating the switch into the control path.

Operations, maintenance and references

Startup

Connect AC, confirm the standby LED, then flip PS_ON to bring up the main rails.

Shutdown

PS_ON off, rear switch off, unplug AC, then wait 30 seconds before touching load wiring.

Fuse replacement

With AC removed, unscrew the front holders and replace only with 5A fast-acting 5×20mm glass fuses.

References

youtu.be/lqjbFdLXqzc

youtu.be/SDymqPkvnt8

youtu.be/n_A-jkpjpcM